

Nipun Arora

M.Sc. Artificial Intelligence | AI Engineer

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Erlangen, Germany

PROFESSIONAL PROFILE

An AI graduate with 3.8 years of experience as an AI Engineer at the **Max Planck Institute** where I built LLM pipelines and agentic workflows across 12+ projects from scratch. I have deep interest in applying AI in Robotics and recently I made an Autonomous Tracking Robot integrating Agentic AI and Robotics.

CORE TECHNICAL SKILLS

- **AI & Deep Learning:** PyTorch, TensorFlow, Computer Vision, LLMs, Foundation Models (VLM/VLA), LoRA.
- **Edge AI & Robotics:** ROS 2, GStreamer, Jetson Xavier/Nano, Raspberry Pi, Arduino.
- **Agentic AI:** LangGraph, Multi-agent pipelines, RAG, MCP, Claude Code, OpenClaw, Hermes-Agent.
- **Software Engineering:** Python, C++, Docker, Git, CI/CD, HPC.

PROFESSIONAL EXPERIENCE

- **AI Engineer** 08/2022 – 03/2026
Max Planck Institute for Human Development - Part Time Berlin, Germany
 - AI engineer within the Center for Humans and Machines (CHM) research group at MPI, contributing across 12+ projects and translating ideas into working AI systems.
 - Tiktok Project: Designed and built a large-scale pipeline incorporating automated data collection, decision making, and web-interactions mimicking human behavior, published in *EPJ Data Science*, 2026.
 - Fine-tuned and deployed LLMs for text classification, topic modelling, and structured data transformation.
 - Developed Agentic AI pipelines to automate data extraction, analysis, and exploration, reducing manual research overhead by 90% and significantly accelerating results.
- **Deep Learning Engineer** 11/2021 – 03/2022
SynergyLabs Technology Gurugram, India
 - Deployed ultra lightweight object detection models on NVIDIA Jetson Nano, NX and Raspberry Pi for real-time inference on Indian National Highways using NVIDIA DeepStream and GStreamer video pipelines.
 - Built and shipped 3 computer vision systems: Automatic Traffic Counter & Classifier, Video Incident Detection, and Automatic Number Plate Recognition (ANPR).

PROJECTS & ACHIEVEMENTS

- **Autonomous Tracking Robot** 03/2026
ROS 2, Raspberry Pi, Arduino, Edge AI, Agentic AI
 - Built a fully autonomous tracking robot controlled by an AI Agent.
 - Integrated Object Detection and Vision Language Models for dynamic context understanding and target pursuit via a custom vision-to-motor control pipeline.
- **Agentic CV Maker** 03/2026
LLMs, Prompt Engineering, LaTeX
 - Built an agentic pipeline that parses CVs and job descriptions to generate tailored LaTeX CVs, cover letters, and interview guides across Gemini, ChatGPT, and Claude backends.

- **Neuro-SAM – Foundation Model for 3D Neural Segmentation** 03/2025 – 09/2025
Anki Lab, AIBE, FAU · Master Thesis
 - Fine-tuned and benchmarked SOTA vision foundation models using LoRA adapters for segmentation of neural dendrites and dendritic spines from 3D microscopy data; trained on FAU's HPC cluster.
 - Developed and released Neuro-SAM as an open-source Python package for fine-tuning and segmenting neural microscopy data.
- **Multi-Modal Quality Monitoring System** 05/2024 – 08/2024
FAPS Lab, FAU · PyTorch Lightning, W&B, HPC
 - Built a configurable multi-modal deep learning pipeline applying sensor fusion of image and time-series sensor data for quality classification of electric motors.
- **Hexbug Tracking Olympiad - 1st Place** 2023
FAU Seminar · YOLO, SAM, Keypoint RCNN, PyTorch
 - Built a multi-stage computer vision pipeline for real-time hexbug tracking: YOLO for object detection, SAM for zero-shot segmentation, and a fine-tuned Keypoint RCNN for precise head localization.
 - Ranked 1st place in the FAU Hexbug Tracking Olympiad.

EDUCATION

- **Master of Science in Artificial Intelligence (M.Sc.)** 05/2022 – 03/2026
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany GPA: 1.8
 - **Thesis:** Neuro-SAM
- **Bachelor of Technology in Information Technology (B.Tech.)** 07/2017 – 06/2021
Guru Gobind Singh Indraprastha University, India GPA: 8.32

PUBLICATIONS

- **Dynamics of Algorithmic Content Amplification on TikTok** 02/2026
Fabian Baumann, Nipun Arora, Iyad Rahwan, Agnieszka Czaplicka EPJ Data Science, Springer Nature

ADDITIONAL INFORMATION

- **Languages:** Hindi (Native), English (C1 – Highly Proficient), learning German
- **Awards:** 1st Place: Hackoverflow ×2 (IBA AG, 2024 & 2025), Hexbug Tracking Olympiad (FAU, 2023)
- **Interests:** Team Sports, Cooking, 3D Printing & Design, DIY, Gaming, Learning Piano